ODSC AI EAST 2026

Data Science & Machine Learning & MLOps

FROM DATA TO DECISIONS:

A HANDS-ON INTRODUCTION TO OPTIMAL DECISION MAKING WITH GUROBI

CAROLINE WEINBERG
Manager, Optimization Support

Gurobi®

SUMMER PURSCHKE
Technical Account Manager

From Predictions to Decisions

- You know what's likely to happen
  - How much demand to expect
  - Which customers might churn
  - What outcomes are likely

Simulate and compare KPIs

Prompt an agent

Predict another value

Follow a standard procedure

Create a network graph

Gurobi

© Gurobi Optimization, LLC. All rights reserved. Confidential.

From Predictions to Decisions

- You know what's likely to happen
  - How much demand to expect
  - Which customers might churn
  - What outcomes are likely
- None of these approaches guarantee the best set of decisions
  - Don’t scale as complexity of decisions grows
  - Break easily when goals or requirements change

Simulation

Generative AI

Machine Learning

Rules

Graphs

So, What’s Missing?

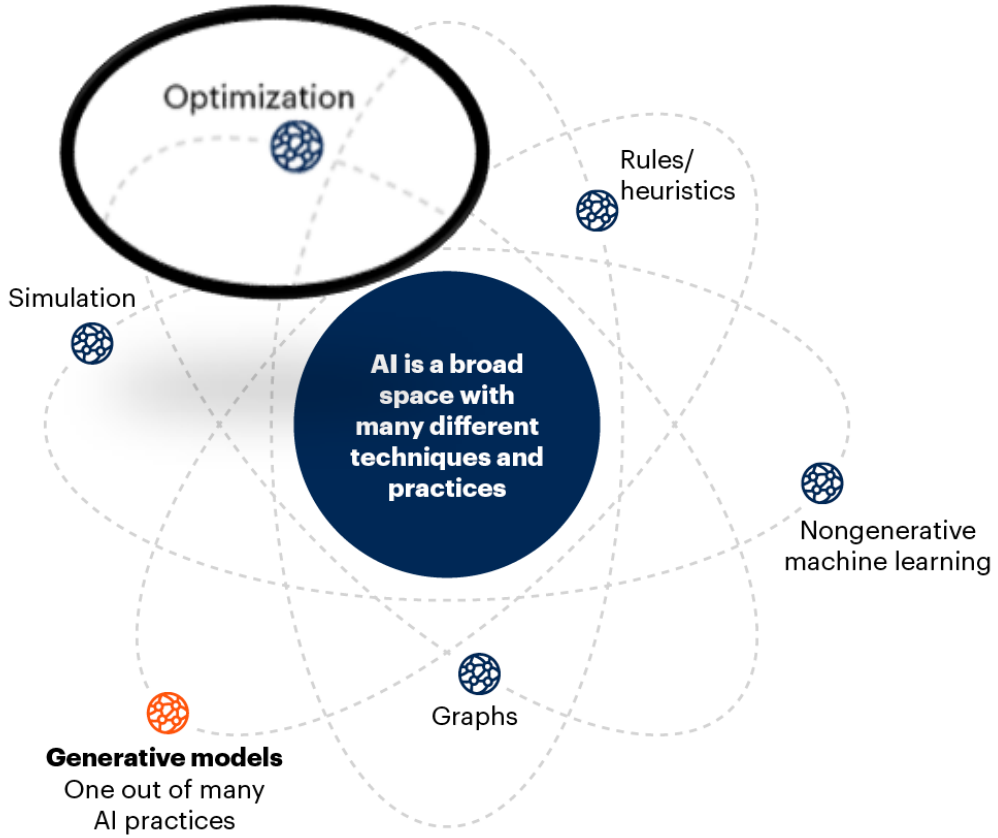
Gurobi

© Gurobi Optimization, LLC. All rights reserved. Confidential.

Optimization in the AI Ecosystem

Optimization is a key component of the AI ecosystem

AI Does Not Revolve Around GenAI



Source: Gartner
806238_C

Gartner

When Not to Use Generative AI
19 March 2024

Gurobi

© Gurobi Optimization, LLC. All rights reserved. Confidential.

Where Optimization Excels

AI Techniques Heat Map

AI technique suitability: Low (L) | Medium (M) | High (H)

Use-case families	Common AI techniques: Generative models	Nongenerative machine learning	Optimization	Simulation	Rules/ heuristics	Graphs
Prediction/ forecasting	L	H	L	H	M	L
Planning	L	L	H	M	M	H
Decision intelligence	L	M	H	H	H	M
Autonomous systems	L	M	H	M	M	L
Segmentation/ classification	M	H	L	L	H	H
Recommendation systems	M	H	M	L	M	H
Perception	M	H	L	L	L	L
Intelligent automation	M	H	L	L	H	M
Anomaly detection/ monitoring	M	H	L	M	M	H
Content generation	H	L	L	H	L	L
Conversational user interfaces	H	H	L	L	M	H
Knowledge discovery	H	M	L	L	M	H

Source: Gartner
806238_C

Gurobi

When Not to Use Generative AI
19 March 2024

Gartner

5

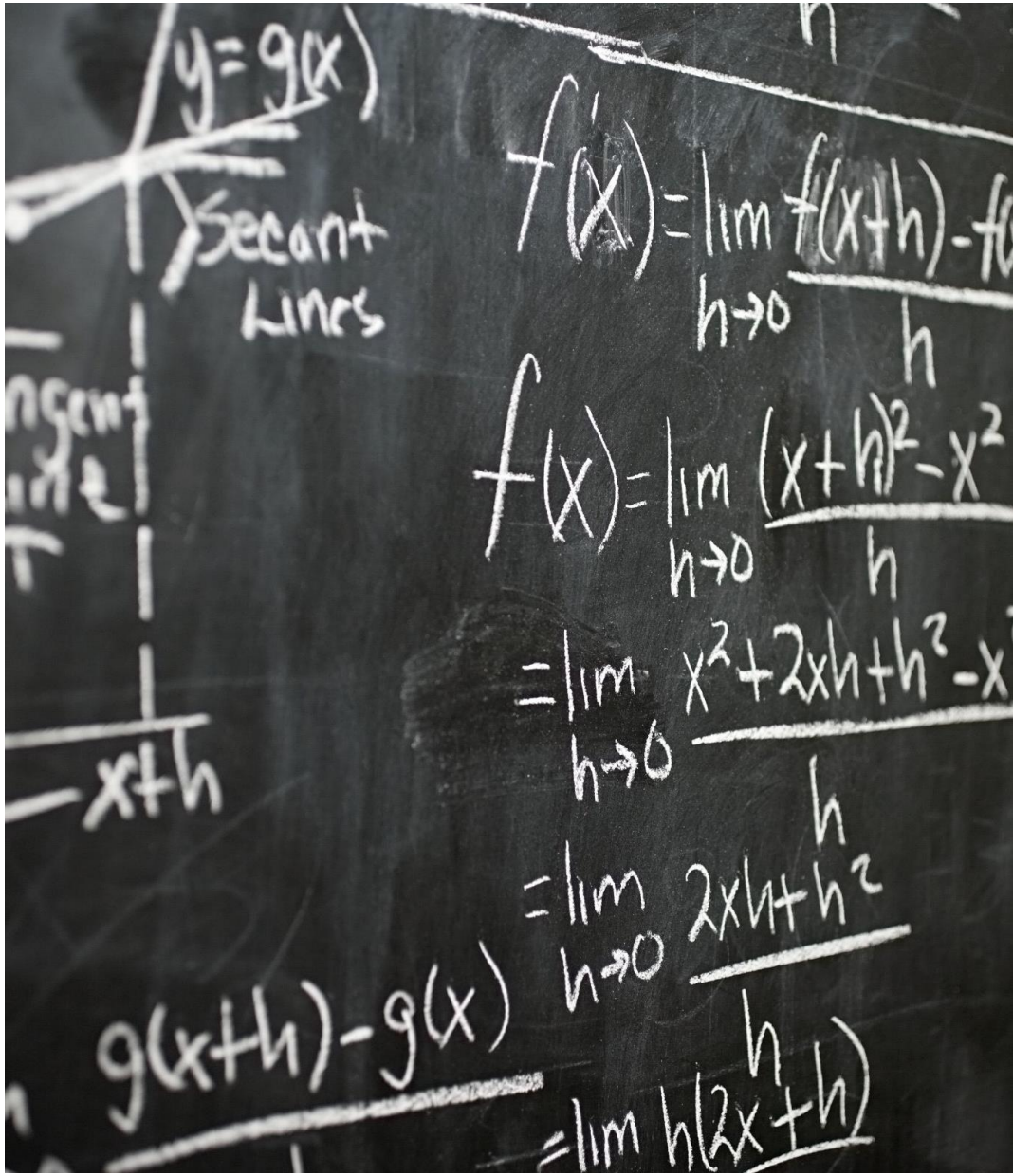
© Gurobi Optimization, LLC. All rights reserved. Confidential.

Mathematical Optimization

A ***prescriptive model*** that explicitly focuses on your decisions, constraints, and objective

Why it matters:

- Produces provably optimal solutions
- Quantifies trade-offs (cost, time, resource limits)
- Easily perform “what if” analyses
- Structure that leads to easier explainability



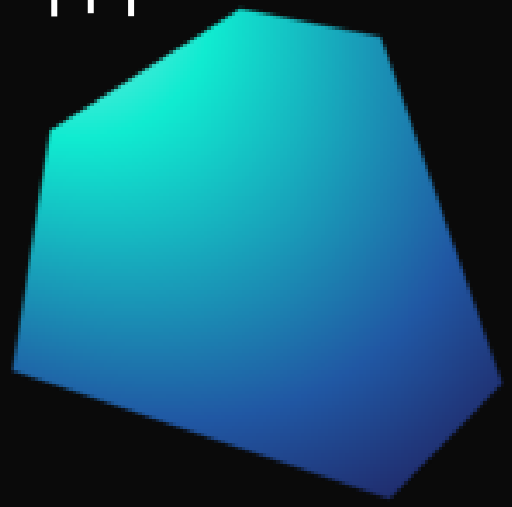
Gurobi

© Gurobi Optimization, LLC. All rights reserved. Confidential.

Your Goals

Your Constraints

Your Questions



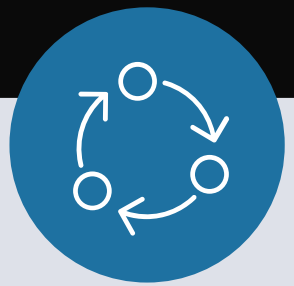
How It Works

Solving real-life problems

A detailed action plan with specific answers to your questions

Understanding the Difference

Mathematical Optimization and the “Optimizer” or “Solver”



Mathematical Optimization

A **method** for modeling your business problem.

Optimizer (Solver)

The **tool** that solves your mathematical optimization model

Problem Statement

Decisions

Goals

Business rules

"Determine how many EV chargers of each type to place at a specific location to maximize the number of vehicles served per day, respecting the resource limitations for budget, power, space, and equipment."

Gurobi

© Gurobi Optimization, LLC. All rights reserved. Confidential.

Mathematical Model Formulation

Decisions

Decision Variables:

$$x_i \in \mathbb{Z}_{\geq 0} \ \forall_i \in I \ (\#\ chargers\ of\ each\ type)$$

Goals

Objective:

$$max \sum_{i \epsilon I} v_i * x_i$$

Business rules

Subject to:

$$\sum_{i \epsilon I} c_i * x_i \leq B \ (budget)$$

$$\sum_{i \epsilon I} p_i * x_i \leq P \ (power)$$

$$\sum_{i \epsilon I} s_i * x_i \leq S \ (space)$$

$$\sum_{i \epsilon I} e_i * x_i \leq E \ (equipment)$$

I: set of charger types
x_i: number of chargers of type i
v_i: vehicles served per day by type i
c_i: cost of type i
p_i: power usage of charger i
s_i: space requirement of charger i
e_i: equipment requirement
B, P, S, E: total available budget, power, space, equipment

© Gurobi Optimization, LLC. All rights reserved. Confidential.

Python Model

```
# Create a model object
model = gp.Model(”EV Charging Network")
```

Decisions

```
# Variables
x = model.addVars(chargers, vtype=GRB.INTEGER, name=”chargers")
```

Goals

```
# Objective
model.setObjective(gp.quicksum(x[i] * vehicles_per_day[i] for i in chargers), sense=GRB.MAXIMIZE)
```

Business rules

```
# Constraints
model.addConstrs((gp.quicksum(x[i] * requirements[i, j] for i in chargers) <= available[j] for j in resources),
name="resource usage")
```

```
# Optimize the model
model.optimize()
```

Gurobi

© Gurobi Optimization, LLC. All rights reserved. Confidential.

Time to get hands-on!

Gurobi

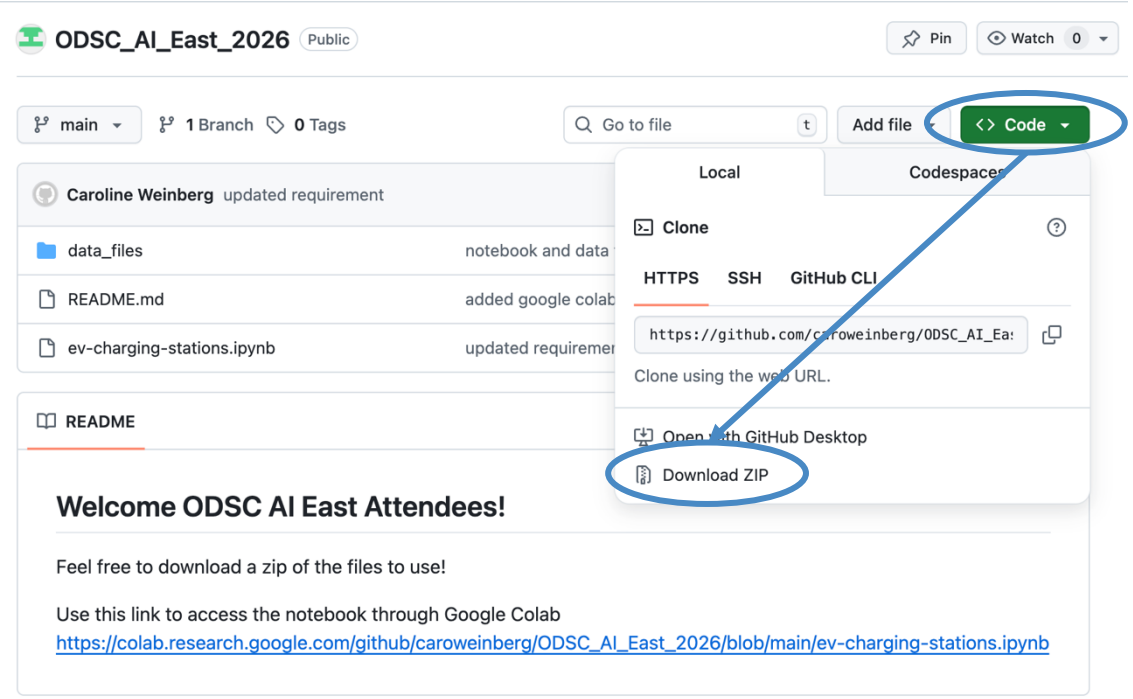
© Gurobi Optimization, LLC. All rights reserved. Confidential.

How to Get Started

1. On your laptop, navigate to

   https://tinyurl.com/gurobiODSC

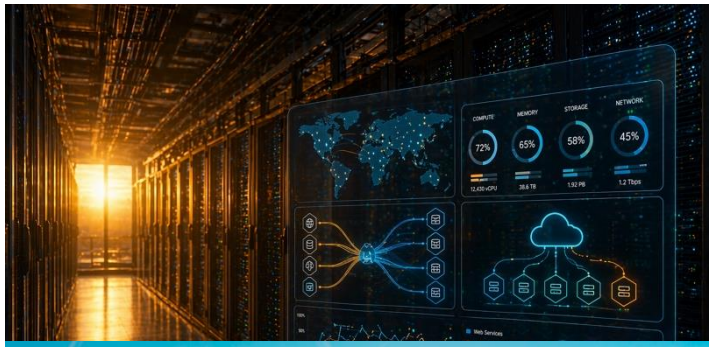
2. Download the ZIP file or navigate to the Google Colab



Gurobi

© Gurobi Optimization, LLC. All rights reserved. Confidential.

Optimization- Silently Running the World



Optimize Power Grid Operations

Balance real-time supply and demand across energy networks to minimize costs and maintain reliability

Optimize Telecommunications Networks

Design network layouts and route traffic to maximize bandwidth, minimize latency, and reduce infrastructure deployment costs.

Optimize Airline Crew Scheduling

Assign pilots and flight crews to schedules that minimize disruptions and costs while complying with union, regulatory, and safety constraints.

Optimize Cloud Computing Resource Allocation

Assign jobs to servers and data centers to maximize performance and efficiency while minimizing cloud infrastructure and energy costs.

Gurobi

© Gurobi Optimization, LLC. All rights reserved. Confidential.

National Football League (NFL)

Challenges

To create the most exciting—and most fair—matchup schedule possible. But they also need to consider factors like broadcast rights, free-agents, away-game limitations, and more. This results in an astronomical number of possible schedules.

Optimized Outcomes

With Gurobi, the NFL scheduling team has increased its decision-making capacity by 2000%.1 It can quickly identify the best possible match-ups, in the time slots with the widest audience—driving revenue while delighting audiences.

"Switching to, and working with, the team from Gurobi has been a great experience. Their solver has met our high expectations, and the support they provide has been fantastic."

Michael North, Sr. Director, NFL

Gurobi

© Gurobi Optimization, LLC. All rights reserved. Confidential.

Blue Yonder

Challenges

To identify optimal price points for entire product lines, subject to complex rules and product relations

Optimized Outcomes

With Gurobi, BlueYonder can now make real-time price adjustments—leading to a 5% increase in product sales and a 20% reduction in inventory.

"With Gurobi, we are able to determine optimal price points efficiently for entire product lines, even subject to complex rules and product relations."

Dr. Ansgar Thiede, Product Manager for Price Optimization, Blue Yonder GmbH

© Gurobi Optimization, LLC. All rights reserved. Confidential.

Machine Learning & Optimization

Gurobi

© Gurobi Optimization, LLC. All rights reserved. Confidential.

Why combine ML & Optimization?

70% of organizations relying solely on ML for AI initiatives will spend more money per solution than those leveraging multiple AI techniques.

Source: Gartner®, Go Beyond Machine Learning and Leverage Other AI Approaches, 26 September 2023. GARTNER is a registered trademark and service mark of Gartner, Inc. and/or its affiliates in the U.S. and internationally and is used herein with permission. All rights reserved.

Gurobi

© Gurobi Optimization, LLC. All rights reserved. Confidential.

Machine Learning + Optimization

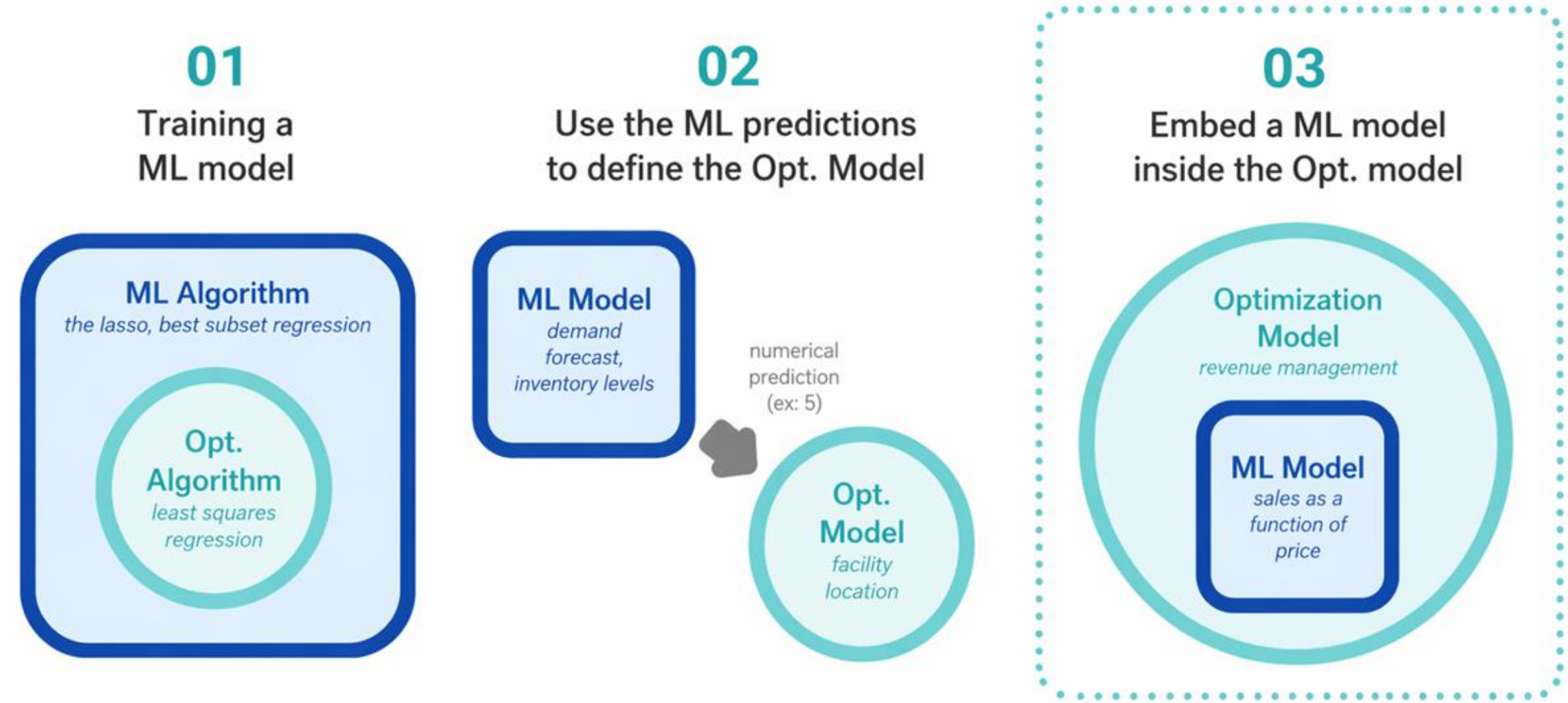
Use ML to:

- Predict customer propensity to buy more with target offers
- Predict price movements before the market does
- Predict supply chain issues
- Predict machine failure

Use Optimization to:

- Decide which discounts to offer to which customers to maximize profits
- Decide how to allocate capital across all investment vehicles
- Decide the least-costly way to reroute shipments
- Decide when to stop production for maintenance to minimize costs and production disruptions

Combining Machine Learning and Optimization



Gurobi

© Gurobi Optimization, LLC. All rights reserved. Confidential.

Gurobi Machine Learning

An open-source Python package to embed trained regression models in a gurobipy model to be solved with Gurobi

- Documentation:
  https://gurobi-machinelearning.readthedocs.io/
- Apache License 2.0
- pip install gurobi-machinelearning
- Supported only on a good-will basis, not through usual Gurobi support
  But we will certainly do our best!
- Example usage: Price optimization with competing products

scikit learn PyTorch dmlc XGBoost Keras LightGBM

© Gurobi Optimization, LLC. All rights reserved. Confidential.

Generative AI & Optimization

Generative AI can democratize access to knowledge and skills

Gurobi

© Gurobi Optimization, LLC. All rights reserved. Confidential.

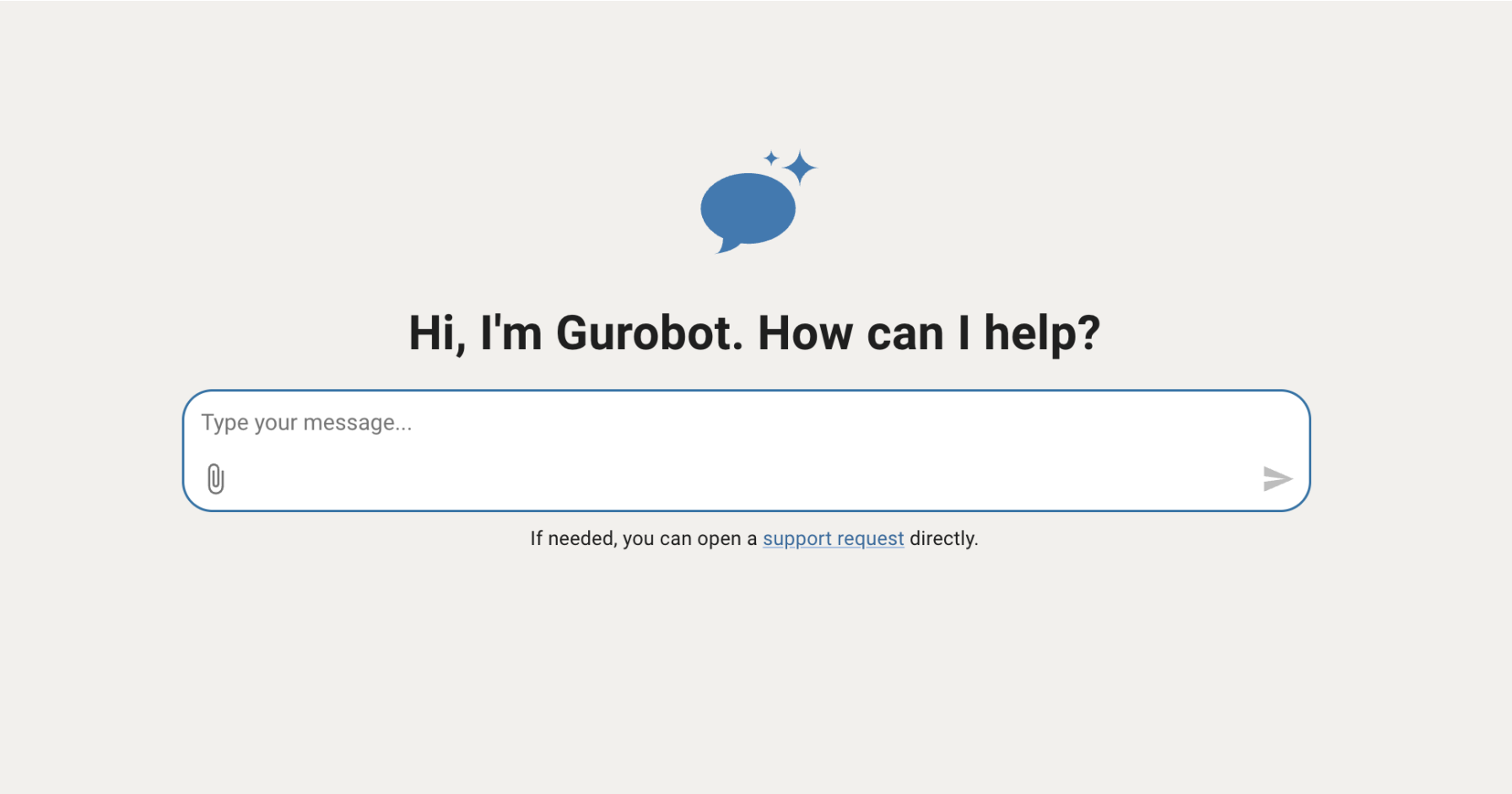
Gurobot

- **Built on our Knowledge:**
  - 500+ knowledge base articles written by PhD Experts
  - Documentation
  - Code Examples
  - Use Cases
  - Modeling Examples

- **Backed by our Experts**
  - Customers can easily convert Gurobot conversations to tickets with our Expert Team
  - Experts will continue developing content to make Gurobot even smarter

© Gurobi Optimization, LLC. All rights reserved. Confidential.

Gurobot



© Gurobi Optimization, LLC. All rights reserved. Confidential.

Gurobot can help us formulate & code our EV problem!

I am part of a municipal parking authority that wants to deploy an EV charging network. We need to decide how many charging stations of each type to install. Five types of charging stations can be installed:

Level1 (120V; 1.4 kW DC): 4 vehicles/day
Level2 (240V; 7.2 kW DC): 10 vehicles/day
Fast50 (50kW DC): 40 vehicles/day
Fast150 (150kW DC): 70 vehicles/day
Ultra350 (350kW DC): 120 vehicles/day

I am providing the following data files:

1. chargers_data.csv: Contains characteristics and performance data for each charger type.

2. resource_available.csv: Contains the total amount of each resource available for the deployment.

3. requirements.csv: Contains the resource consumption requirements for each charger type. This is a multi-index table showing how much of each resource is needed to install one unit of each charger type.

We need to decide whether or not to install each of the above charger types, and how many of each.
We want to minimize total cost, ensuring that the number of vehicles serviced per day is at least 90% of the total possible vehicles serviced if that were our only goal. There is a fixed cost for each type of charger installed, and a variable cost for each charger of that type installed. I have some additional business requirements to satisfy:

- Resource limits: Total usage of budget, power, space, and equipment cannot exceed what's available
- Category minimums: Must have at least 10 fast chargers (any type) and at least 15 standard chargers (any type)

Can you help me formulate and generate the code to solve this problem?

© Gurobi Optimization, LLC. All rights reserved. Confidential.

Gurobot shows its thinking process

Gurobot

- The user wants me to generate Python code for the EV charging network deployment problem. Let me follow the guidelines:

  1. Search the knowledge base for relevant information about mixed-integer programming, fixed costs, and best practices
  2. Ask the coding-advisor tool for best practices in Python
  3. Generate clean, well-documented code following best practices
  4. The code should:
     - Read the three CSV files
     - Calculate V_max (maximum vehicles serviced)
     - Build the cost minimization model
     - Include all constraints
     - Solve and display results

  Let me start by searching the knowledge base and getting coding best practices.

- Querying the knowledge base.

- Getting best practices for Python.

Gurobi

© Gurobi Optimization, LLC. All rights reserved. Confidential.

Gurobot provides a formulation & code

Gurobot

Perfect! Now let me provide you with a comprehensive formulation for your EV charging network deployment problem.

Optimization Problem Formulation: EV Charging Network Deployment

---

Context

You are a municipal parking authority planning to deploy an optimal mix of electric vehicle (EV) charging stations across five technology levels. Each charger type has different installation costs (fixed and variable), resource requirements (budget, power capacity, physical space, equipment), and service capacity (vehicles per day). The goal is to minimize total deployment cost while meeting service level targets and operational constraints, including resource availability, minimum deployment requirements for fast and standard

Gurobi

© Gurobi Optimization, LLC. All rights reserved. Confidential.

Gurobot provides a formulation & code

ev_charging_optimizer.py

COPY DOWNLOAD

```python
"""
EV Charging Network Deployment Optimizer

This module solves the optimization problem for deploying an EV charging network
with multiple charger types, minimizing total cost while meeting service level
and operational constraints.
"""

import gurobipy as gp
from gurobipy import GRB
import pandas as pd
import os


class EVChargingOptimizer:
    """Optimizer for EV charging network deployment decisions."""

    def __init__(self, env):
        """
        Initialize the EV Charging Optimizer.

        Args:
            env: Gurobi environment
        """
        self.env = env
        self.model = None

        # Data attributes
```

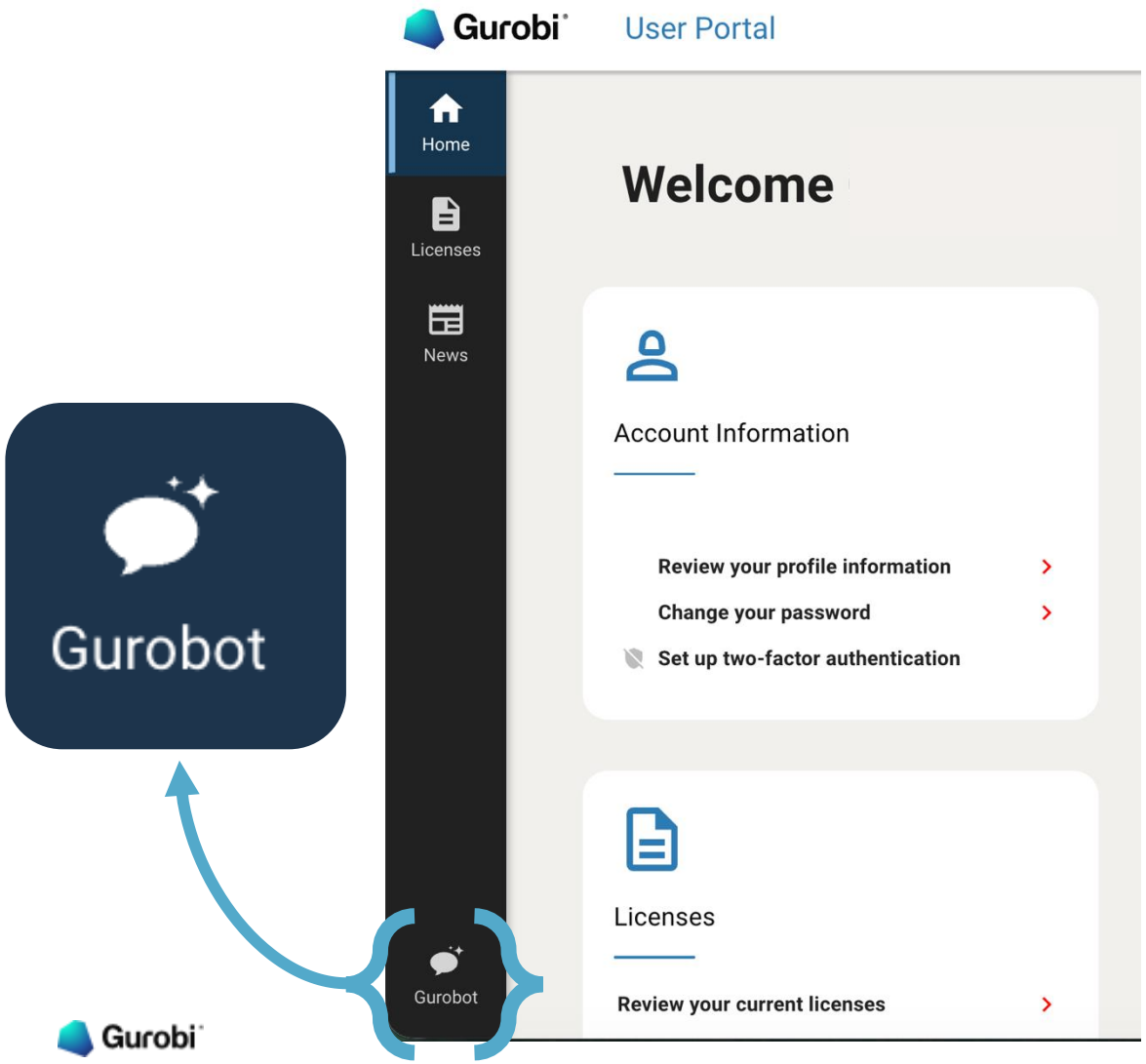
ev_charging_optimizer.log

COPY DOWNLOAD

```log
==================================================================
STEP 3: Solving Optimization Model
==================================================================

Gurobi Optimizer version 13.0.1 build v13.0.1rc0 (linux64 - "Amazon Linux 2023.10.20260302")

CPU model: Intel(R) Xeon(R) Processor @ 2.50GHz, instruction set [SSE2|AVX|AVX2]
Thread count: 2 physical cores, 2 logical processors, using up to 2 threads

Non-default parameters:
TimeLimit  300

Optimize a model with 16 rows, 10 columns and 48 nonzeros (Min)
Model fingerprint: 0xe054bff1
Model has 10 linear objective coefficients
Variable types: 0 continuous, 10 integer (5 binary)
Coefficient statistics:
  Matrix range     [5e-01, 2e+04]
  Objective range  [2e+02, 2e+04]
  Bounds range     [1e+00, 1e+00]
  RHS range        [1e+01, 2e+05]

Presolve removed 5 rows and 2 columns
Presolve time: 0.00s
Presolved: 11 rows, 8 columns, 32 nonzeros
Variable types: 0 continuous, 8 integer (3 binary)
Found heuristic solution: objective 48800.000000
Found heuristic solution: objective 48400.000000
```


© Gurobi Optimization, LLC. All rights reserved. Confidential.

Accessing Gurobot



Gurobi Portal Account

- If you do not yet have a Gurobi Portal Account, register

  https://portal.gurobi.com/iam/register/

- Use your company or academic email address
- Once registered, log in with your username and password
- Click on the Gurobot icon in the bottom left corner and start chatting!

© Gurobi Optimization, LLC. All rights reserved. Confidential.